

■■ LUMINA — Complete Evaluator & Demo Script

AI-Driven Crime Intelligence & Analytical Platform (CIAP) · KSP Datathon 2026

■■ Demonstration Flow & Timing (Total: 5–7 Minutes)

Phase	Module / Topic	Duration	Core Objective
01	The Hook & Problem Statement	45s	Excel silo bottleneck → Real-time AI Intelligence
02	Tactical Command Hub & GIS	90s	ST-DBSCAN Hotspots, Patrol Routing, Multi-base Maps
03	Global Search & Voice Briefings	90s	Sub-50ms search, Pinned Dossiers, Neural Kannada TTS
04	Strategic Analytics & Risk Scores	60s	Repeat offender index (>85), Statewide KPI breakdown
05	Criminal Network Topology	45s	Multi-accused syndicates, Cytoscape graph hubs
06	AI Copilot (Bilingual RAG)	60s	Gemini 2.5 Flash with live DB context & zero hallucinations
07	Official PDF Briefing & Config	45s	One-click sealed KSP Dossier & voice speed presets

Phase 1: The Hook & Introduction (0:00 - 0:45)

Visual: Start on the Command Hub (') with the dark glassmorphic interface loaded.

"Respected judges and senior officers: Today, law enforcement faces a critical bottleneck. Across 31 districts and 209 police stations in Karnataka, thousands of FIRs and suspect records are stored in siloed tabular formats. When an incident occurs, identifying repeat offenders, active crime corridors, or multi-jurisdiction syndicates takes hours of manual dossier searches.

Introducing **LUMINA** — an AI-powered Crime Intelligence and Spatiotemporal Analytics Platform built natively for the Karnataka State Police on Zoho Catalyst. Lumina transforms raw case records into sub-second predictive intelligence, visual graph topology, and automated tactical patrol dispatch."

Phase 2: Tactical Command Center & Hotspot GIS (0:45 - 2:15)

Actions: Pan across Karnataka map (5,005 FIRs). Click the Indiranagar corridor cluster to expand the Tactical Patrol Dispatch card. Switch map styles (Esri Dark → Midnight → Satellite).

"On the primary Command Hub, Lumina runs real-time ST-DBSCAN spatiotemporal clustering across every active FIR. Rather than static heatmaps, our algorithm groups incidents by geographic radius and temporal density.

Notice this critical cluster in the Indiranagar–MG Road corridor: with a single click, Lumina calculates the threat score (94/100), isolates the predominant crime group (Cybercrime & Extortion), and generates an automated tactical patrol route — recommending **Patrol Alpha-4** with an estimated response time of 6 minutes.

Officers can toggle instantly between Esri Dark night operations, tactical midnight, and high-resolution satellite

imagery."

Phase 3: Global Search & Bilingual Voice Briefings (2:15 - 3:45)

Actions: Press Ctrl + K. Search '1693'. Click ■ Pin to add to Priority Dossiers. Click ■■ Audio Intelligence Briefing. Toggle to Kannada (■■■■■) and play.

"Pressing Ctrl + K launches our Universal Search Intelligence modal. Lumina searches across case numbers, accused aliases, victim profiles, IPC sections, and narrative text.

Let's look up FIR #1693/2026. Immediately, the full intelligence brief loads: primary accused, threat level, and station jurisdiction. Watch as I click 'Play Audio Briefing' — our integrated Google Neural TTS synthesizes a real-time spoken debriefing for officers on patrol.

With one toggle, Lumina translates and vocalizes the briefing in pure Kannada, guaranteeing full regional accessibility for field personnel."

Phase 4: Strategic Analytics & Risk Scoreboard (3:45 - 4:45)

Actions: Navigate to Overview (`/overview`) and Risk Scores (`/risk-scores`). Point out the Repeat Offender Flagging Index (scores ≥ 85).

"Switching to the Overview tab, Lumina provides state-level supervisory situational awareness. We track live status breakdowns — cases under investigation, chargesheeted, and closed.

Under the Risk Scoreboard, our predictive weighting algorithm analyzes criminal history, repeat arrest counts, and violent crime severity to compute a dynamic 0–100 threat score. Any offender exceeding our supervisory threshold (≥ 85) is automatically flagged for surveillance."

Phase 5: Criminal Network Topology (4:45 - 5:30)

Actions: Click Network Topology (`/network`). Click a central repeat offender node to reveal connected accomplices across multiple cases.

"Organized crime rarely operates in isolation. In the Network Topology module, Lumina visualizes multi-accused syndicates. Red nodes represent high-threat offenders, while blue nodes represent specific FIRs.

By clicking any suspect, officers instantly uncover shared accomplices across different police stations that would remain hidden in standard tabular files."

Phase 6: AI Copilot & Ground-Truth RAG Assistant (5:30 - 6:30)

Actions: Open AI Chatbot (`/ai-chatbot`). Query in English, then toggle to Kannada and ask a question. Show live database citations.

"Lumina features a dedicated AI Copilot backed by Gemini 2.5 Flash with strict RAG ground-truth injection. When an officer queries the copilot, it queries the live database first and injects factual context into the system prompt.

It never hallucinates fake FIRs. In Kannada mode, it enforces Karnataka Police terminology, providing seamless bilingual intelligence."

Phase 7: Official PDF Briefing & System Config (6:30 - 7:15)

Actions: Click [■ Export Briefing] on `/overview``. Open the sealed PDF. Open ■■ System Config to show calibrated voice speeds.

"Finally, supervisory officers can click 'Export Briefing' on the overview page to instantly compile an official, sealed Karnataka State Police Intelligence Dossier ready for morning briefings.

In the System Configuration drawer, officers can tune ST-DBSCAN cluster radii, alert chimes, map layers, and calibrate neural speech playback from Ultra Slow (0.75x) to Ultra Fast (1.70x).

Lumina is 100% deployed on Zoho Catalyst serverless infrastructure, scalable, secure, and ready for production deployment across Karnataka State Police. Thank you!"

■ Evaluator Q&A; Cheat Sheet

Evaluator Question	Winning Answer / Technical Rationale
How do you prevent AI hallucinations?	Our RAG pipeline parses FIR references and executes Data Store queries first. Real database rows are injected as immutable ground truth into the Gemini prompt with a strict directive to reject unverified cases.
What is the backend architecture?	100% Zoho Catalyst Serverless (Python 3.13 Advanced I/O <code>`api_service`</code> + Scheduled <code>`etl_cron`</code> + <code>`lumina-client`</code> web hosting) with sub-second execution latency and zero server maintenance overhead.
How does it work in field / patrol conditions?	Lumina includes AES-256 encrypted local state caching, offline-resilient query fallbacks, mobile-responsive tactical layouts, and calibrated neural voice narration for hands-free audio briefings.